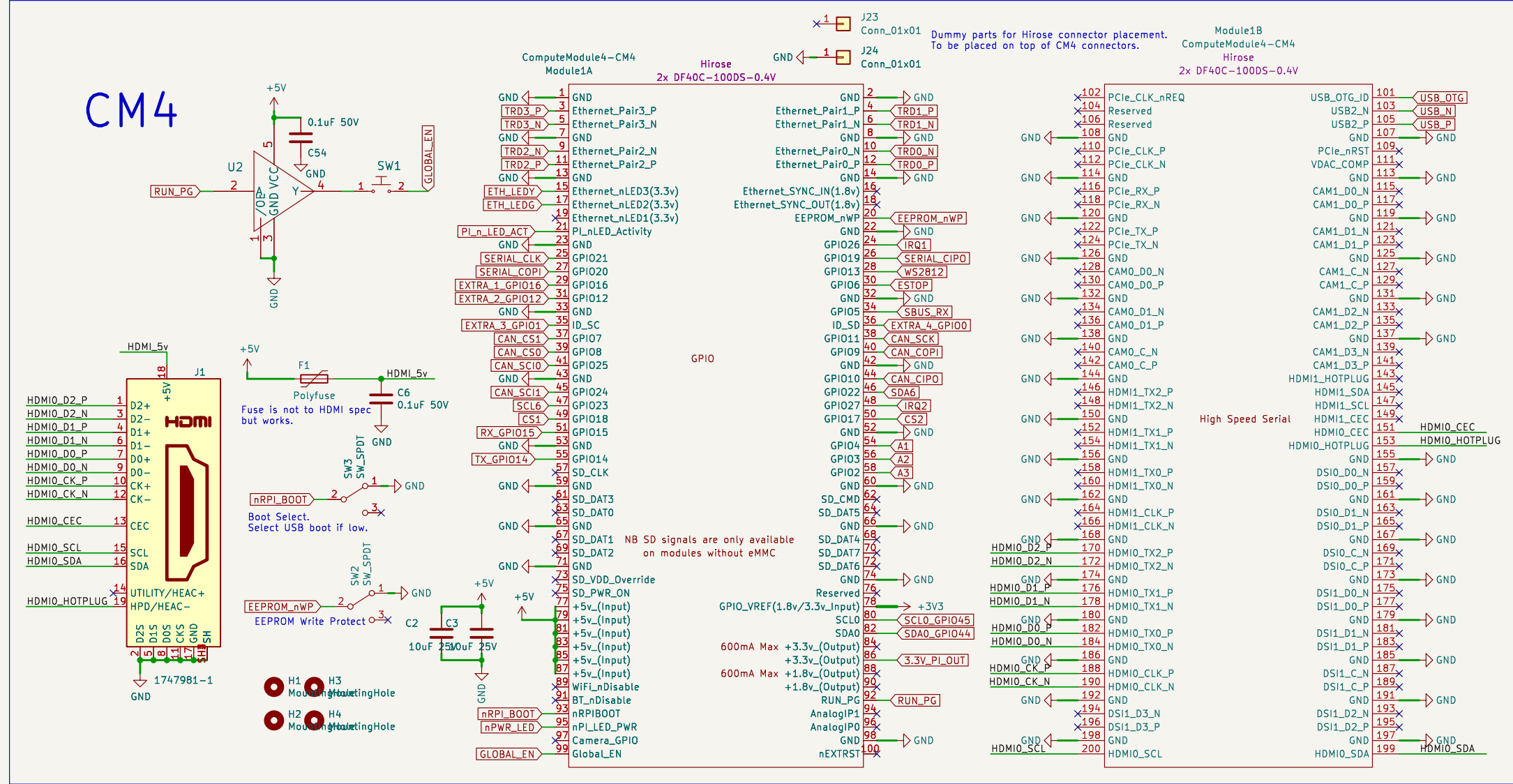
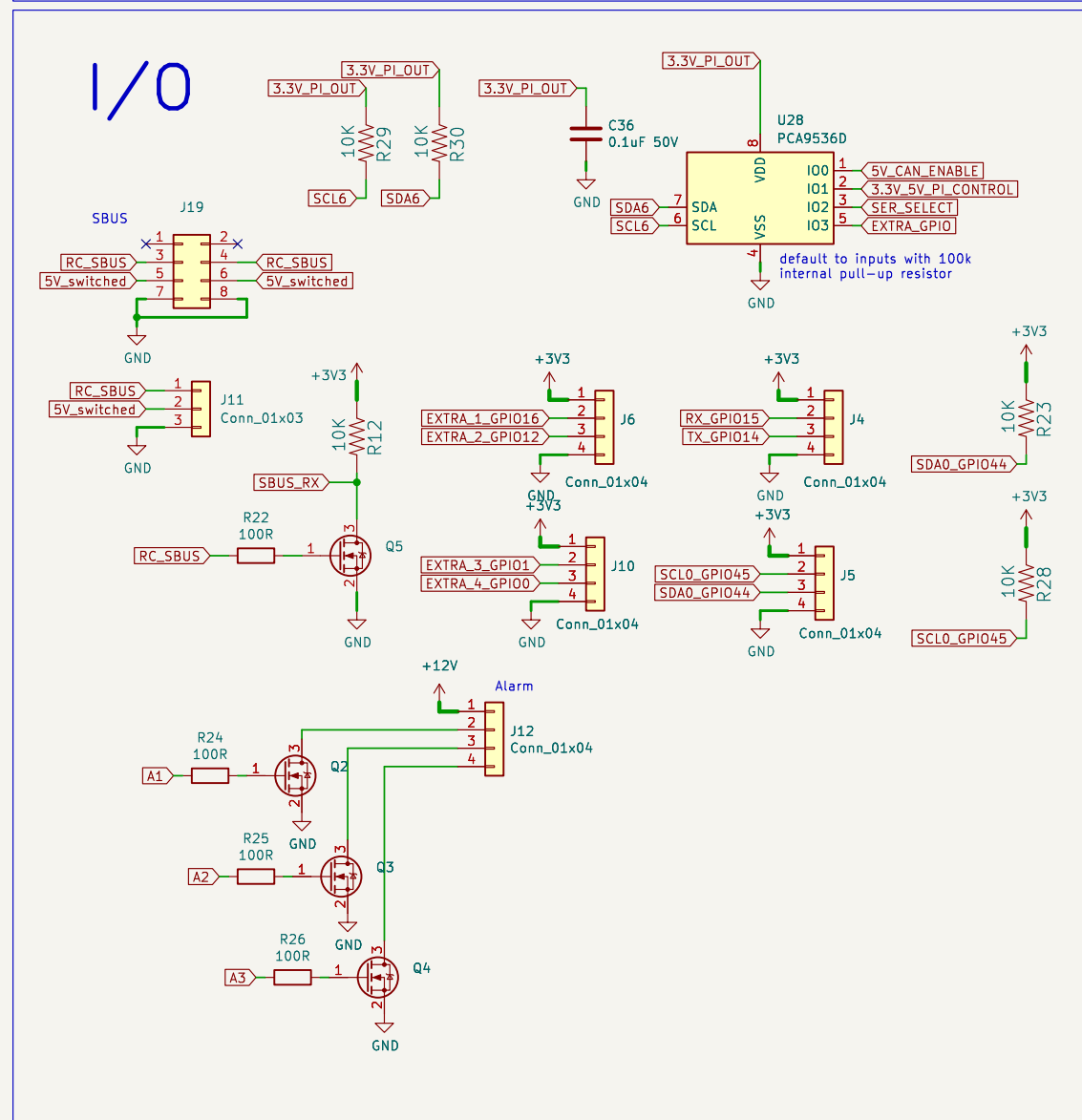


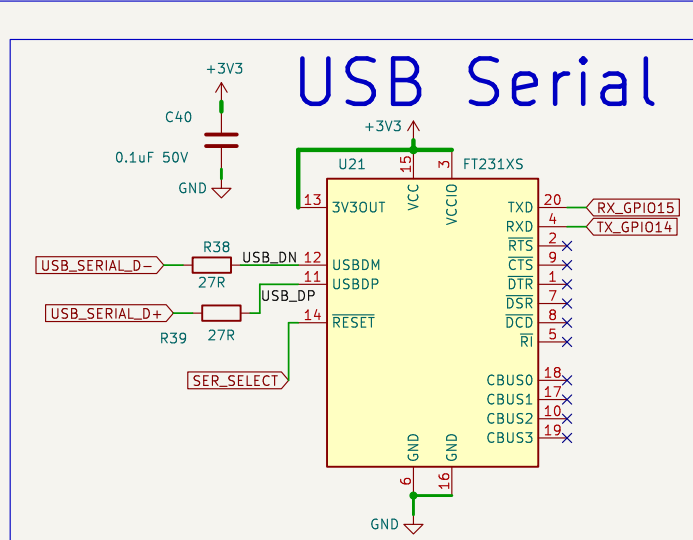
CM4



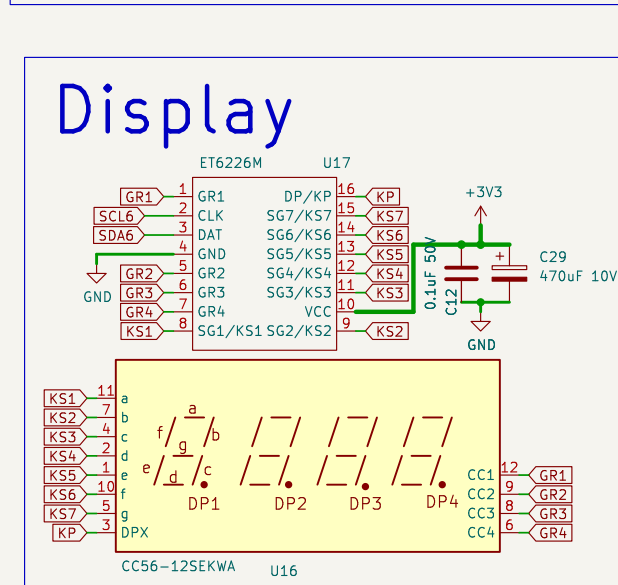
I/O



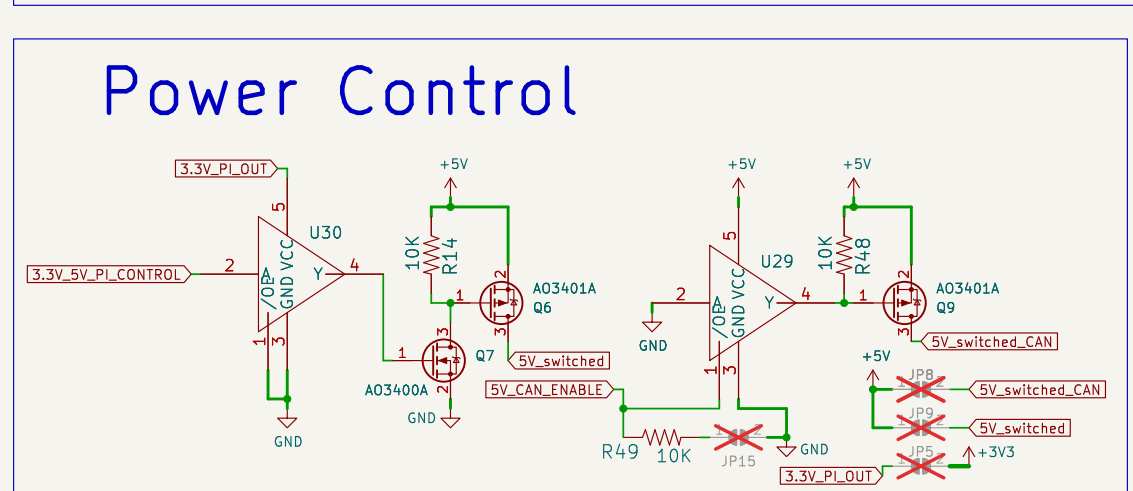
USB Serial



Display

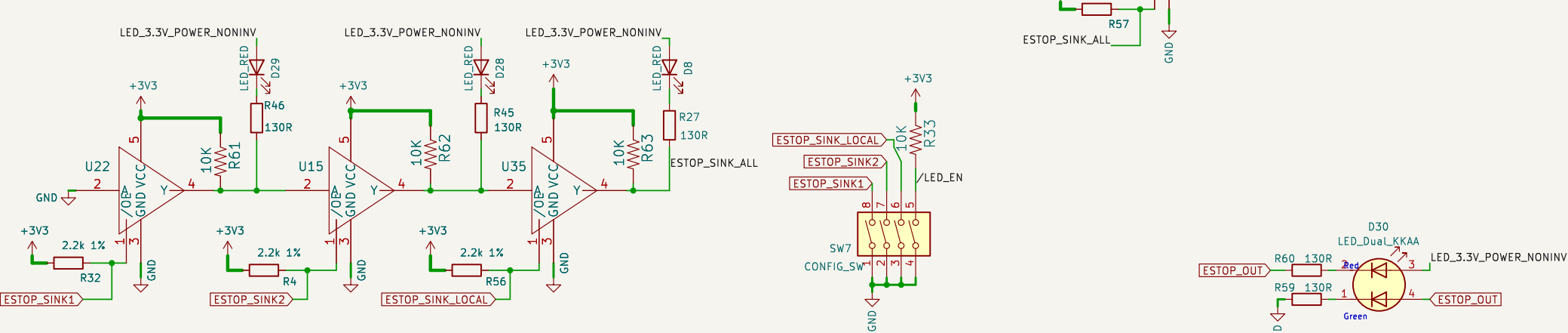


Power Control

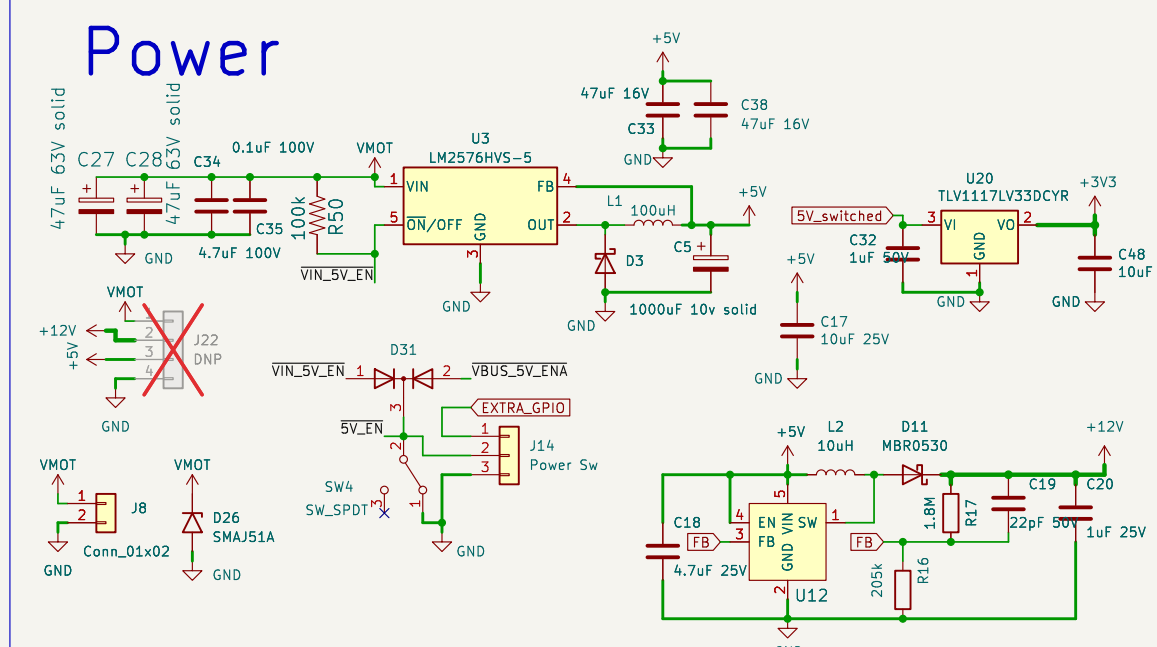


SPI Naming Note:
COP1 "Controller Out, Peripheral In"
CIPO "Controller In, Peripheral Out"
This is a slight deviation from OSWHA who recommend PICO/POCI, but those terms swap the order of the controller compared to previous terms, and I find MISO=CIPO easier to intuit than MISO=POCI.

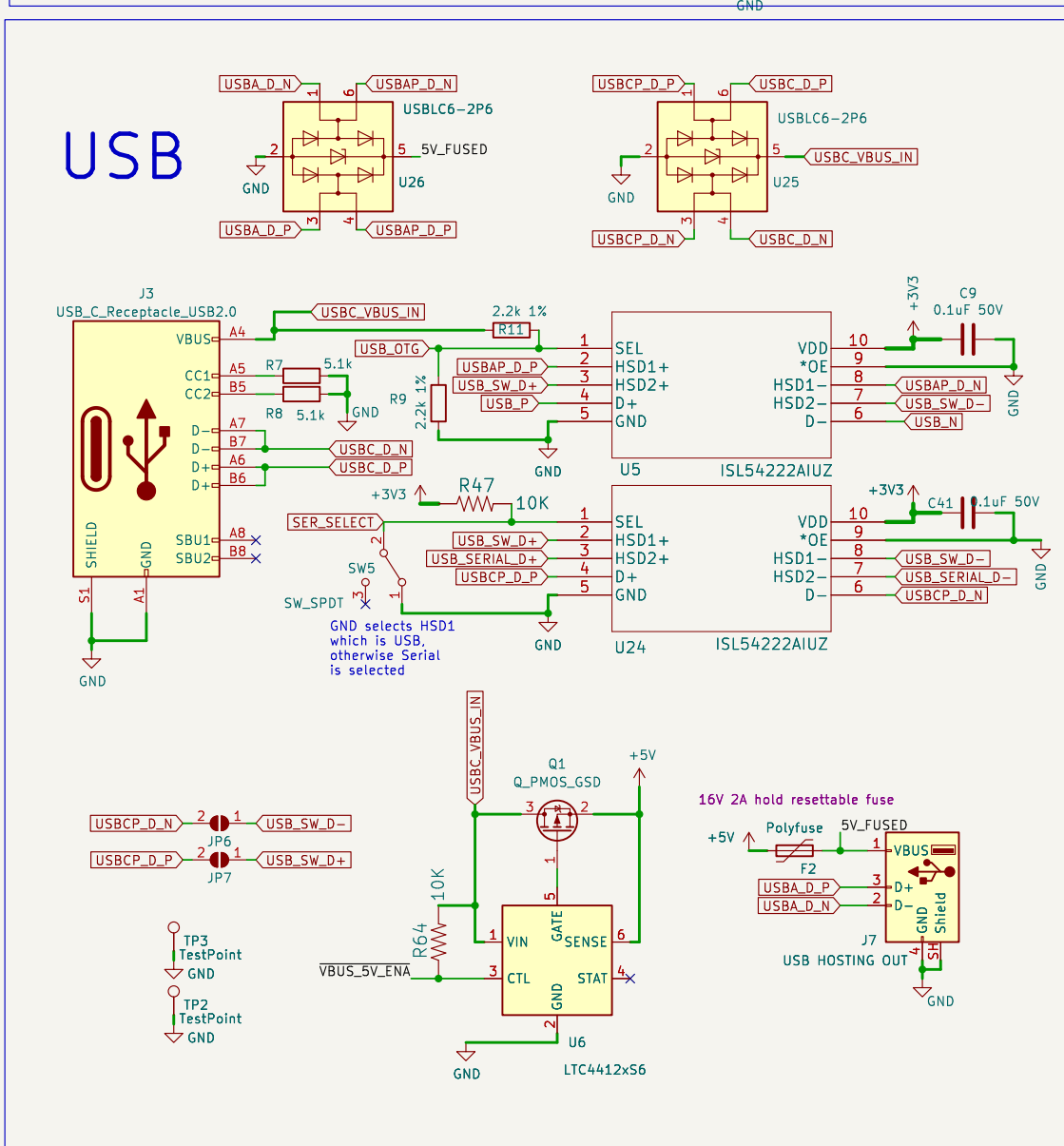
LED Enable and E-Stop Config



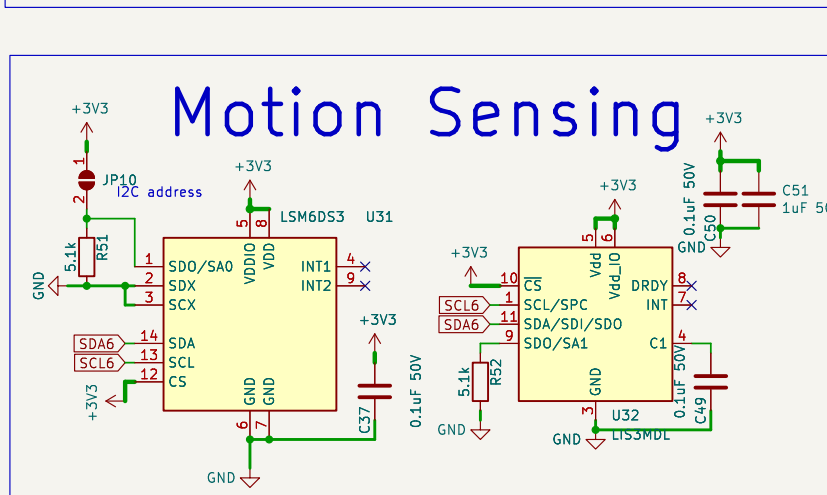
Power



USB



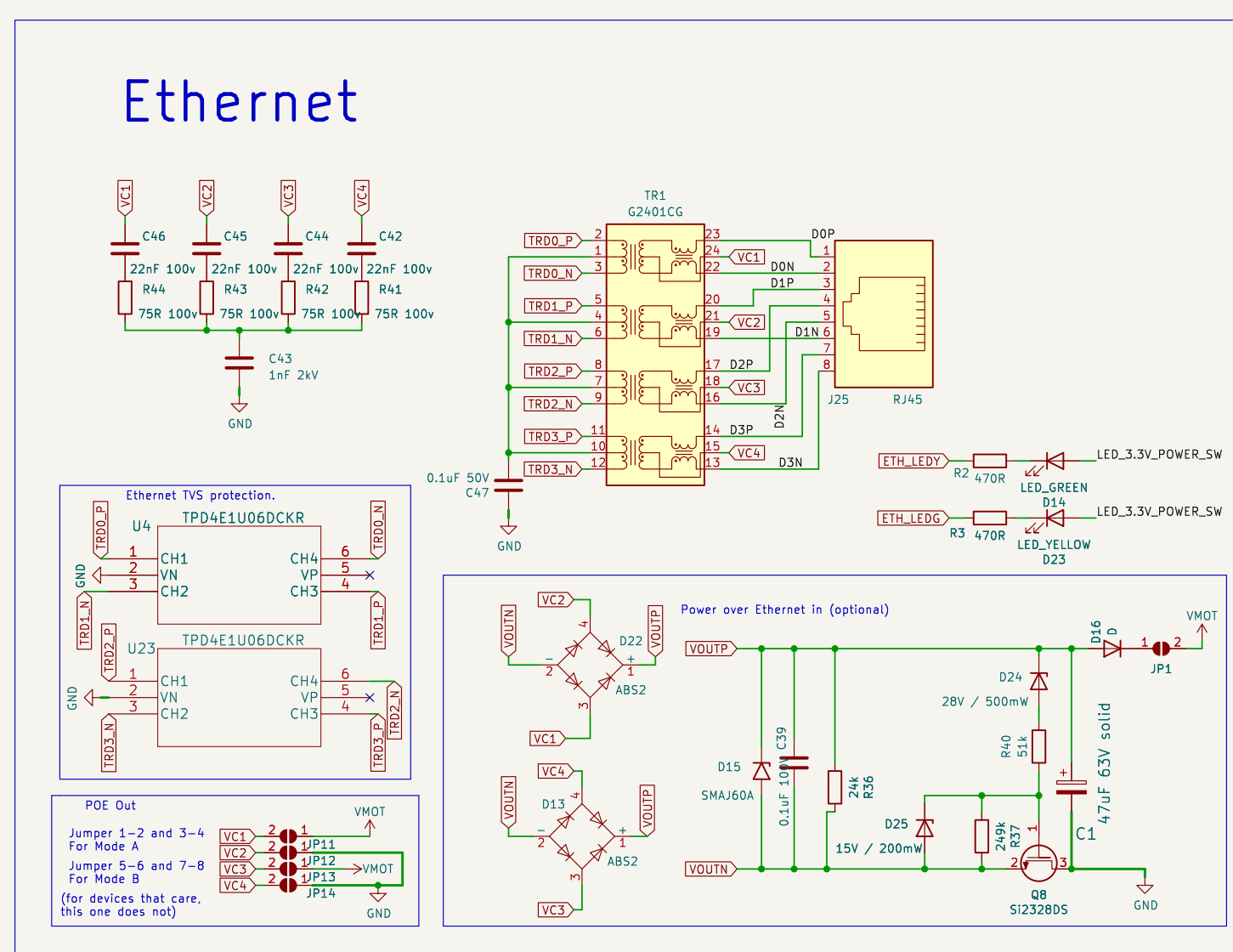
Motion Sensing



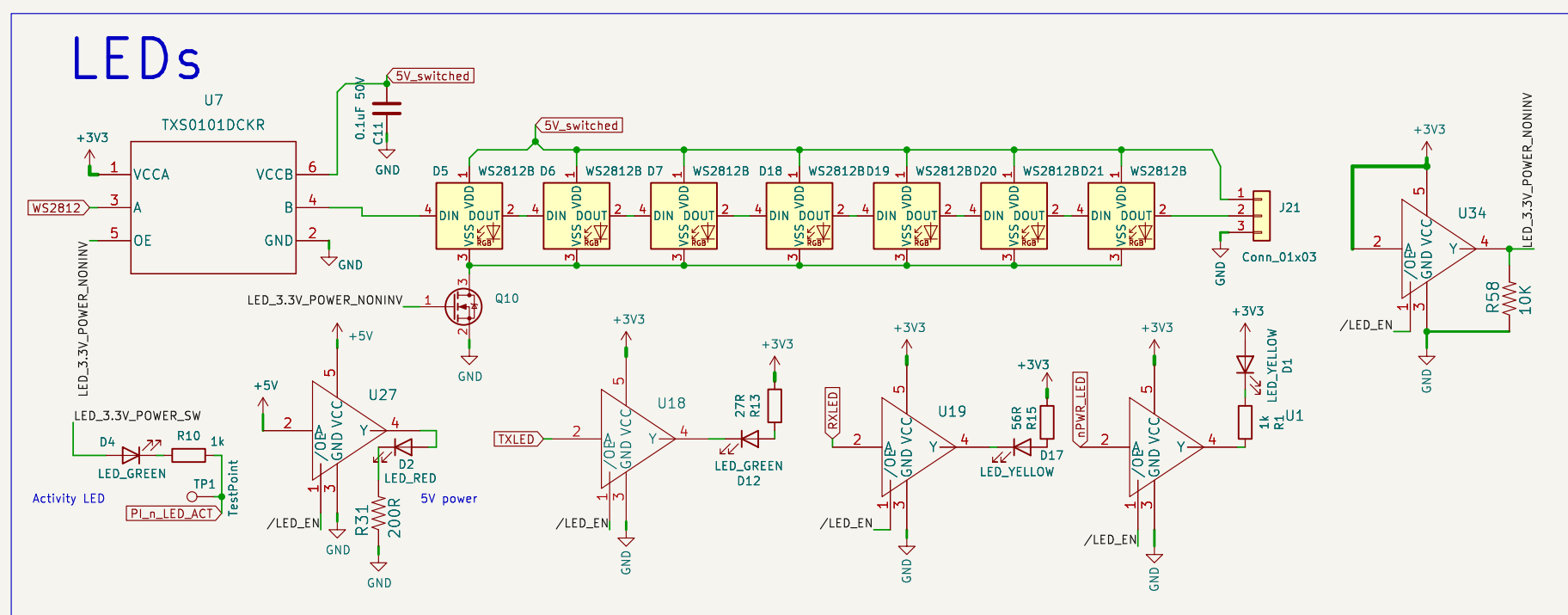
Changelog

– July 10 2024
Expand E-stop circuit to provide indicator lights for each branch
Add local motherboard branch of e-stop circuit
Provide jumper to control whether external 5v over CAN is on or off at startup.
Add extra SBus connector to eliminate cabling.
Add pullup resistor to power switch signals.
Add I2C ADC for input voltage sensing

Ethernet



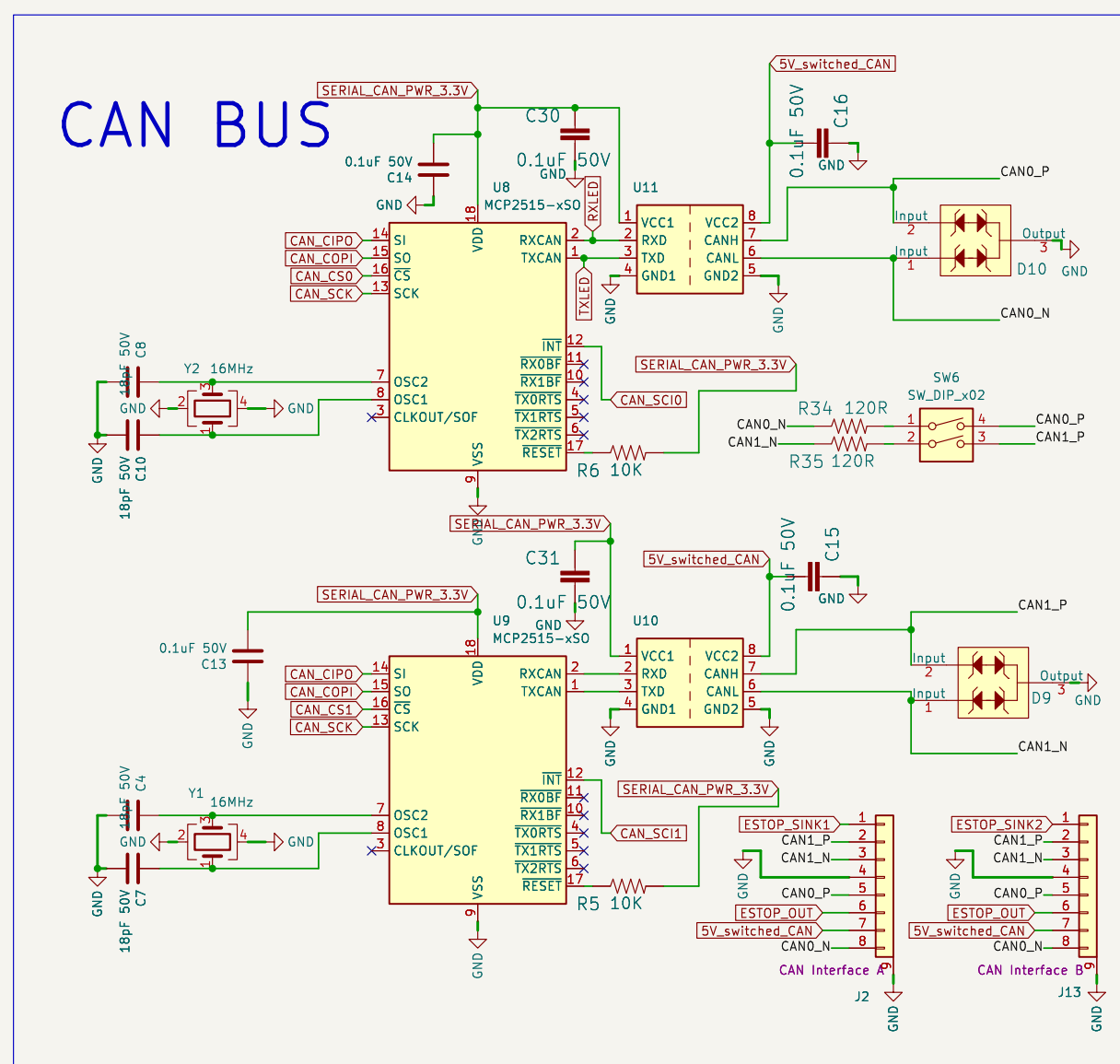
LEDs



3.3V_PL_OUT – always on 3.3v from pi, for GPIO expander
+3.3V – switched 3.3v regulator, fed off 5v rail controlled by GPIO expander
SERIAL_CAN_PWR_3.3V – jumper-selectable rail can feed from either of the two above, controls the serial and CAN chips, which have kernel modules that may or may not get unhappy if their chips reset

We use an external 3.3v regulator because the Pi's onboard 3.3v regulator can fail if shorted, so we use it as little as possible.

CAN BUS



Serial

